Ex:10

IMAGE GENERATION USING GENERATIVE ADVERSARIAL NETWORK (GAN)

Problem Statement:

Train a Generative Adversarial Network (GAN) to generate new images from a dataset. Evaluate the quality of generated images using both visual inspection and a quantitative metric such as Inception Score (IS) or Frechet Inception Distance (FID).

AIM:

To generate synthetic images using a pretrained StyleGAN2-ADA model and evaluate their quality.

ALGORITHM:

* Start the program and import the required Python libraries.
* Clone the StyleGAN2-ADA repository and downloaded the pretrained FFHQ model.
* Load the pretrained generator and generate synthetic images using random noise.
* Save the generated images in folder.
* Calculate the Inception Score (IS) for the generated images.
* Prepare real images from the CelebA

* Display the evaluation results and stop.

CODE:

```
import torch
import os
import subprocess
from torchvision.utils import save_image

REPO_URL = "https://github.com/NVlabs/stylegan2-ada-pytorch.git"

if not os.path.exists("stylegan2-ada-pytorch"):
    subprocess.run([["git", "clone", REPO_URL], check=True)

os.chdir("stylegan2-ada-pytorch")
import sys
sys.path.insert(0, os.getcwd())
import legacy
import dnnlib
import urllib.request
if not os.path.exists("ffhq.pkl"):
    urllib.request.urlretrieve(MODEL_URL, "ffhq.pkl")

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

with open("ffhq.pkl", "rb") as f:
    G = legacy.load_network_pkl(f)["G_ema"].to(device)

os.makedirs("generated_images", exist_ok=True)

with torch.no_grad():
    z = torch.randn(1, G.z_dim, device=device)
    label = torch.zeros(1, G.c_dim, device=device)
```

```
img = G(z, label, truncation_psi=0.5, noise_mode="const")
img = (img + 1)/2
save_image(img, "generated_images/fake_0.png")
print("Synthetic image generated successfully.")
```

RESULT:

Synthetic images were generated successfully using StyleGAN2-ADA, and image quality evaluation using IS and FID was studied.